

Shashaank Reddy Gurralla

Data Engineer

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PROFESSIONAL SUMMARY

Innovative Data Engineer with 6+ years of experience delivering enterprise-scale data platforms across healthcare analytics, financial services, and B2B supply chain domains. Expert in architecting end-to-end ETL/ELT pipelines, medallion architectures, and real-time streaming solutions using Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Snowflake, Microsoft Fabric, Apache Kafka, Apache Airflow, DBT (Data Build Tool), and SSIS (SQL Server Integration Services). Deep hands-on experience modernizing legacy workloads (HDFS, Oozie, YARN, SSIS) to Azure cloud-native architectures, implementing Delta Lake, Unity Catalog, Change Data Capture (CDC), SCD Type 1/2/3, and HIPAA/GDPR/CCPA compliance. Proven expertise in data quality frameworks, data observability, data contracts, and DataOps practices including CI/CD automation, automated testing, and infrastructure as code using Terraform. Strong healthcare domain knowledge in CMS HCC risk adjustment, ICD-10, LOINC, NDC, CPT, and MAO-002 standards. Proficient in PySpark, Python, SQL, T-SQL, PL/SQL, DBT, Apache Airflow, and SSIS with containerized deployments using Docker and Kubernetes.

TECHNICAL SKILLS

Cloud Platforms: Microsoft Azure, Azure Data Factory (ADF), Azure Databricks, Azure Data Lake Storage (ADLS Gen2), Azure Synapse Analytics, Azure Event Hubs, Azure Stream Analytics, Azure Logic Apps, Azure Function Apps, Azure Key Vault, Azure DevOps, Azure Blob Storage, Azure Container Instances, Microsoft Fabric, Snowflake

ETL/ELT & Orchestration: Azure Data Factory (ADF), Apache Airflow (DAGs, Operators, Sensors, XComs, SLA Monitoring), DBT (Data Build Tool), SSIS (SQL Server Integration Services), Apache Oozie, YARN, IBM DataStage

Big Data & Streaming: Apache Spark, PySpark, Delta Lake, Apache Kafka (Consumer Groups, Schema Registry, Schema Evolution), Apache Hive, HDFS, Hadoop, Azure HDInsight, PySpark Structured Streaming

Programming & Scripting: Python, PySpark, SQL, PL/SQL, T-SQL, HiveQL, SnowSQL, Bash/Shell Scripting

Databases & Warehouses: Snowflake, Azure Synapse Analytics (Dedicated SQL Pool), Azure SQL Database, MS SQL Server, Oracle, MongoDB, Cassandra, Cosmos DB

Data Modeling & Transformation: DBT (Data Build Tool), Erwin, ER/Studio, Star Schema, Snowflake Schema, Dimensional Modeling, SCD Type 1/2/3, CDC, Data Contracts

Data Quality & Observability: Great Expectations, Data Contracts, Schema Registry, Data Lineage, SLA Monitoring, Data Validation Frameworks, Data Profiling

Infrastructure & DevOps: Terraform, Docker, Kubernetes, GitHub Actions, Azure DevOps, Git, GitHub, SonarQube, Service Principal OAuth2, REST API, CI/CD

Data Governance: Unity Catalog, RBAC, HIPAA, GDPR, CCPA, Data Lineage, PHI Encryption, Data Cataloging

BI & Reporting: Power BI, Databricks Genie, Azure Synapse Serverless SQL

File Formats: JSON, CSV, Parquet, Avro, ORC, XML, EDI (X12 850/810/856), Flat Files

Project Management: JIRA, Rally, Azure DevOps (ADO), Agile, Scrum, Kanban, SDLC, DataOps

AI/ML Tools: GitHub Copilot, Claude, GPT-4, Databricks Genie, LLM-assisted Development, RAG Pipelines

CERTIFICATIONS

- DP-750 – Microsoft Certified: Azure Databricks Data Engineer Associate
- Academy Accreditation – Databricks Generative AI Fundamentals
- Generative AI for Everyone – DeepLearning.AI

PROFESSIONAL EXPERIENCE

Data Engineer

Client: Optum

Jun 2024 – Present
Remote (San Antonio, Texas)

Business Context: Built and scaled the Prospective Risk Engine, a healthcare risk adjustment analytics platform serving 17 U.S. health plan clients including UnitedHealth Group, Aetna, Humana, and Centene. Engineered rule-based frameworks scanning claims, pharmacy, and clinical data to surface Suspects and Recaptures, routing prioritized worklists to care managers and medical coders.

- Built and managed multi-client data ingestion pipelines using Azure Data Factory (ADF), Azure Databricks, and Snowflake, onboarding diverse source formats (JSON, CSV, Parquet) into ADLS Gen2 raw storage and Delta Lake tables using Python, PySpark, and SQL, implementing RBAC via Unity Catalog to enforce data access controls and restrict offshore access on ACA-specific sensitive data.
- Architected Azure Data Factory (ADF) pipelines using Copy, ForEach, Lookup, Get Metadata, and Web activities to orchestrate full and incremental ETL/ELT data extraction into Delta Lake tables, implementing Change Data Capture (CDC) with SCD Type 1 and SCD Type 2 mechanisms, reducing daily processing time by 70% and enabling scalable, low-latency data ingestion.
- Developed and maintained DBT (Data Build Tool) models for gold layer transformation logic, writing modular SQL-based DBT models with incremental materializations, DBT tests (schema tests, data tests), DBT sources, and DBT documentation to enforce data quality and lineage tracking across the medallion architecture, replacing legacy transformation logic with testable, version-controlled DBT pipelines.

- Orchestrated end-to-end data pipeline workflows using Apache Airflow, designing Airflow DAGs with task dependencies, PythonOperator, BashOperator, DatabricksRunNowOperator, and sensor-based triggers to coordinate Azure Data Factory (ADF) pipeline execution, DBT model runs, Databricks notebook jobs, and Snowflake data loads with alerting and retry logic.
- Migrated legacy Oozie-based Python codebase to Azure Databricks with Spark-compatible PySpark transformations, replacing traditional Python UDFs with native Spark functions and achieving a 30% improvement in processing speed across ETL/ELT pipelines.
- Implemented medallion architecture (Raw-Bronze-Silver-Gold) with PySpark, DBT (Data Build Tool), and SQL-based data quality validation frameworks using Great Expectations for automated data profiling, schema validation, and expectation suites, promoting only quality-passed records through each layer and enforcing data contracts between producer and consumer teams.
- Delivered analytics-ready gold layer datasets to Azure HDInsight (Hadoop/YARN) and Azure Databricks clusters, managing HDInsight cluster operations via REST API calls, supporting parallel analytics workloads processing 500GB+ daily across multiple concurrent clients using PySpark, Apache Hive, and YARN.
- Leveraged Databricks LTS clusters, Databricks Jobs, and Databricks Notebooks to orchestrate overnight data loading workflows, configuring scheduler-based, file arrival, and table update triggers with idempotent pipeline design ensuring exactly-once processing semantics and SLA compliance across all client workloads.
- Designed automated CI/CD deployment pipelines using GitHub Actions with YAML workflows and REST API integration with Azure Databricks, leveraging Service Principal OAuth2 authentication and Terraform for infrastructure provisioning to deploy across dev, staging, and production environments, achieving SonarQube code coverage above 90%.
- Provisioned and managed Azure Synapse Analytics (Dedicated SQL Pool) as a centralized data warehouse, writing optimized SQL pipelines to aggregate and expose gold layer datasets for enterprise reporting, Power BI dashboards, and cross-client analytics, reducing average query latency by 45%.
- Architected a post-analytics storage layer using Azure Synapse Analytics Dedicated SQL Pool, integrating Azure Data Factory (ADF) pipelines and Databricks jobs to load processed analytics output from gold layer into Azure Synapse, enabling faster query execution and enterprise-wide reporting.
- Applied healthcare domain knowledge including Claims (ICD-10), Lab (LOINC), Pharmacy (NDC), CPT, MAO-002, and CMS risk adjustment standards with ICD-to-HCC mappings, ensuring HIPAA, GDPR, and CCPA compliance and encrypting PHI data at rest and in transit.
- Migrated Prospective Risk Engine to Microsoft Fabric, replacing ADLS Gen2, ADF, and Databricks with Fabric Lakehouse, Fabric Data Factory, and Spark Notebooks under OneLake, consolidating storage, compute, and governance into one platform.
- Adopted AI-assisted development using GitHub Copilot, Claude, and GPT-4 to deploy two production PySpark frameworks within 2 months – Loss Gain Analysis and Monthly Analytics Report – reducing QA validation time by 60% through automated statistical summaries.
- Utilized AI/LLM models to auto-generate Python unit test cases for ETL/ELT repositories, improving SonarQube code coverage above 90% and ensuring consistent quality gate compliance across the codebase.
- Utilized Databricks Genie for natural language querying over Gold layer datasets, reducing dependency on manual SQL reporting and empowering non-technical stakeholders to self-serve analytics.

Data Engineer

Jul 2022 – May 2024

Client: Capital One

Normal, Illinois

- Migrated large volumes of customer data from on-premises SQL Server, NoSQL databases (MongoDB, Cassandra), Oracle, and file formats (JSON, XML, Avro, Parquet) to ADLS Gen2 and Azure Synapse Analytics using Azure Data Factory (ADF) with Self-Hosted Integration Runtime (SHIR), leveraging ADF Connectors, Linked Services, and ODBC/JDBC drivers.
- Configured and executed ETL/ELT workflows using Python, PySpark, PL/SQL, and SSIS (SQL Server Integration Services), extracting, transforming, and loading data into ADLS Gen2, Azure SQL Database, and Azure Synapse Analytics, migrating legacy on-premises Oracle and SSIS ETL processes to Azure cloud.
- Developed and maintained SSIS (SQL Server Integration Services) packages for legacy data integration, designing SSIS Control Flow and Data Flow tasks, Execute SQL Tasks, Flat File Sources, and OLE DB Destinations, managing SSIS package deployment using SSIS Catalog (SSISDB) and SQL Server Agent job scheduling before migrating workloads to Azure Data Factory (ADF) and Apache Airflow.
- Built Apache Airflow DAGs for orchestrating multi-step ETL/ELT pipelines, using Airflow PythonOperator, BashOperator, SqlSensor, and ExternalTaskSensor to coordinate data extraction, PySpark transformations on Databricks, DBT model runs, and data loading into Snowflake and Azure Synapse Analytics, with alerting, retry, and SLA monitoring.
- Implemented DBT (Data Build Tool) transformation pipelines on Snowflake, building modular DBT models with ref() and source() macros, DBT incremental materializations, DBT snapshot models for SCD Type 2, and DBT tests (unique, not_null, relationships, accepted_values), generating DBT documentation and lineage graphs for cross-team data contract enforcement.
- Implemented Delta Lake architecture on Azure Databricks providing ACID transactions, data versioning, SCD Type 1/2/3, surrogate key assignment, and CDC, delivering pipeline reliability and efficient data storage and processing.
- Managed Snowflake Multi-Cluster Warehouses for query performance at scale, writing complex SnowSQL queries for ETL operations with optimized partitioning, leveraging Snowflake Time Travel for historical data access, auditing, and compliance.
- Designed Star Schema and Snowflake Schema data models using Erwin for Azure Synapse Analytics Dedicated SQL Pool, supporting dimensional modeling and optimized query performance for Power BI reporting.
- Leveraged Azure Event Hubs and Apache Kafka for real-time event streaming, configuring Kafka consumer groups, topic partitioning, offset management, and Confluent Schema Registry for schema evolution, ensuring near-zero latency and minimal data loss across high-throughput streaming pipelines using PySpark Structured Streaming.
- Containerized PySpark ETL/ELT applications using Docker and deployed to Azure Kubernetes Service (AKS) for scalable, reproducible pipeline execution, managing container orchestration, resource allocation, and auto-scaling policies across development and production environments.
- Designed REST APIs following RESTful principles and OpenAPI specifications, integrating with Apache Airflow DAGs for end-to-end

pipeline orchestration across diverse data sources and destinations.

- Implemented data quality validation frameworks using Great Expectations, defining expectation suites for schema validation, null checks, referential integrity, and statistical distribution checks across ingested datasets, integrating quality gates into Apache Airflow DAGs to halt pipeline execution on data contract violations.
- Automated CI/CD processes using GitHub Actions and Azure DevOps with Terraform for infrastructure-as-code provisioning of Databricks workspaces, Snowflake warehouses, and Azure resources, implementing test-driven development pipelines for source code management and seamless deployment.
- Utilized Azure Blob Storage lifecycle management to automate data tiering across Hot, Cool, and Archive tiers, optimizing cloud storage costs by 35% across large-scale datasets.

Data Engineer

Jun 2020 – Dec 2021

Client: OpenText

Chennai, India

- Engineered Azure Data Factory (ADF) pipelines to ingest B2B supply chain transaction data including EDI formats (X12 850, 810, 856) from trading partners into ADLS Gen2, using PySpark to transform raw EDI flat files into structured Parquet format for downstream analytics and reconciliation reporting.
- Developed SSIS (SQL Server Integration Services) packages for batch data integration of legacy supply chain and vendor master data, designing SSIS Data Flow pipelines with Lookup, Derived Column, Conditional Split, and OLE DB Destination transformations, deploying to SSIS Catalog (SSISDB) with SQL Server Agent scheduling for automated nightly batch loads.
- Built DBT (Data Build Tool) models on Snowflake for transforming vendor performance, lead time, and inventory turnover datasets, using DBT incremental models, DBT macros, and DBT tests to build reproducible, version-controlled transformation layers aligned with medallion architecture principles.
- Designed Apache Airflow DAGs to orchestrate end-to-end supply chain data pipelines, coordinating ADLS Gen2 file arrival sensors, ADF pipeline triggers, PySpark transformation jobs, SSIS package execution, DBT model runs, and Snowflake data loads using Airflow task dependencies, retries, and SLA monitoring.
- Designed Azure SQL Database schemas for vendor master data, purchase orders, and shipment tracking, writing T-SQL stored procedures, CTEs, and window functions to automate reconciliation logic between supplier confirmations and order management systems.
- Integrated Snowflake as analytical warehouse for supply chain reporting, loading cleansed datasets using SnowSQL and Python automation, building SCD Type 2 pipelines for historical tracking of supplier and product master data changes over time.
- Designed multi-layer Snowflake schemas with raw, cleansed, and aggregated layers mirroring medallion architecture principles, implementing RBAC at schema and table level using SQL-based grants and role assignments to enforce data access boundaries across teams.
- Built Azure Event Hubs-based security log ingestion pipelines with Apache Kafka consumer groups, processing high-throughput security event streams using PySpark Structured Streaming with watermarking and windowed aggregations, landing enriched records into ADLS Gen2 for real-time threat detection and compliance auditing.
- Implemented data quality validation checks using Great Expectations integrated into Apache Airflow DAGs, defining expectation suites for schema enforcement, null detection, and referential integrity across ingested supply chain datasets, ensuring data contract compliance before downstream consumption.
- Containerized Python-based data ingestion microservices using Docker, deploying to Azure Container Instances for scalable file monitoring and event-driven pipeline triggering, detecting new file arrivals in ADLS Gen2 and automatically invoking downstream Azure Data Factory (ADF) pipelines.
- Provisioned and managed cloud infrastructure using Terraform scripts for Azure Data Factory, ADLS Gen2 storage accounts, Azure SQL Database, and Azure Event Hubs, implementing infrastructure-as-code practices for reproducible, version-controlled environment provisioning across dev, staging, and production.
- Developed Python REST API clients for third-party vendor data extraction, handling OAuth2 authentication, pagination, rate limiting, and retry logic with idempotent request handling, landing raw API responses into ADLS Gen2 for downstream PySpark-based transformation and processing.
- Implemented end-to-end data observability and monitoring using Azure Monitor, Log Analytics, and custom PySpark-based data profiling jobs, tracking pipeline SLA compliance, data freshness, row count anomalies, and schema drift alerts to ensure proactive issue detection across all production data pipelines.

EDUCATION

Master of Science in Computer Science – Illinois State University

Bachelor of Science in Computer Science Engineering (Data Analytics) – Vellore Institute of Technology